Comparative Performance of Lunar Earth-Departure Systems

Vehicle Mass, Propulsion Capability, and Mission Architecture

Rob Tompkins

October 13, 2026

Abstract

This study compares Apollo's integrated S-IVB/CSM/LM stack with SLS/ICPS–Orion, Starship HLS, and Blue Moon Mark II lunar transportation concepts. Assembly-specific mass and propellant ledgers distinguish low Earth orbit delivery, translunar injection, lunar arrival, landing, and return. Apollo postflight data provide a historical reference; unpublished modern vehicle inventories are represented by explicit numerical scenarios and sensitivity ranges rather than asserted flight specifications. The analysis compares single-launch and separate-launch mission complexity while distinguishing mission completion from crew survival. It examines the phase-dependent emergency value of Apollo's attached lunar module, including independent life support and landing propellant usable through its own propulsion system. Storable, pressure-fed hypergolic propulsion reduces selected ignition and feed dependencies, but does not by itself establish greater propellant capacity or overall reliability. Apollo's approximately two-hour low lunar orbit and the approximately 6.5-day near-rectilinear halo orbit are compared through recovery opportunities, resource endurance, and surface access; orbital period alone is not a measure of rescue time. Lander height, width, support geometry, and touchdown dynamics are related to tip-over risk on uneven terrain, with historical Apollo probability estimates distinguished from unvalidated modern comparisons. The study recommends evaluating a new integrated Apollo-type lunar stack and defines preliminary requirements for the proposed Apollo S-VB stage family: a departure-only variant and a larger ascent-and-departure variant. Super Heavy- and SLS-based implementations are outlined with explicit mass, propulsion, and integration constraints. These concepts support further engineering study, not a demonstrated launcher fit or a quantitative claim of superior crew safety.

Contents

1 Purpose and Method **4**

2 Mass Accounting and Rocket Equation **4**

3 Numerical Vehicle Data **4**

3.1 Apollo: S-IVB, CSM, and LM 4

3.2 SLS, ICPS, Orion, and ESM 5

3.3 Starship Human Landing System 6

3.4 Blue Moon Mark II 6

4 Summary Mass Table **7**

5 Architecture-Level Transport Efficiency **7**

6 Normalized Versus Mission-Native Analysis **8**

7 Apollo and Artemis as Different Mass Architectures **8**
7.1 Vehicle Dynamics Beyond Mass Ratio 8
7.1.1 Mass Ratio and Propellant Fraction 8
7.1.2 Specific Impulse and Exhaust Velocity 9
7.1.3 Ideal Delta-V 9
7.1.4 Payload Sensitivity 10
7.1.5 Thrust-to-Weight Ratio 10
7.1.6 Propellant Flow Rate and Burn Duration 10
7.1.7 Ideal Delta-V Versus Effective Delta-V 11
7.1.8 Orbital Energy 11
7.1.9 Translunar Injection 11
7.1.10 Delta-V Margin 12
7.1.11 Staging and Discarded Mass 12
7.1.12 Usable Propellant, Reserves, and Residuals 13
7.1.13 Mission-State Mass and Energy Ledger 13
7.1.14 Assembly-Specific Mission Ledgers 13
7.1.15 Implications for Vehicle Comparison 19

8 Single-Launch and Separate-Launch Architecture Complexity **19**
8.1 A Reliability Model with Separate Outcomes 20

9 Emergency Resources, Reliability, and Crew Survivability **21**
9.1 Apollo's Attached Lunar Module as an Emergency Resource 21
9.2 Chemo-Mechanical Backup: Storable Hypergolic Propulsion 21
9.3 What Can and Cannot Be Credited to Artemis 23
9.4 Emergency Endurance and Recoverability 23

10 Crew Safety: Two-Hour Low Lunar Orbit Versus 6.5-Day NRHO **24**
10.1 Period Is Not Rescue Time 24
10.2 Published NRHO Abort Durations and Their Limits 24
10.3 Safety Implications for Mission Design 25

11 Lander Dimensions, Uneven Terrain, and Tip-Over Probability **26**
11.1 Height, Body Width, and Landing-Gear Footprint 26
11.2 Static Stability on a Slope 26
11.3 Why Touchdown Can Tip a Statically Stable Lander 27
11.4 Estimating the Probability of Tip-Over 28
11.5 Requirements for the Proposed Apollo-Type Lander 29

12 Conclusion: Reconsider an Integrated Apollo-Type Lunar Stack **29**

13 Proposed Implementations: Apollo-Type Upper Stack on Modern Launchers **30**
13.1 Common Lunar Stack and Mission Specification 30
13.2 Apollo S-VB: Preliminary Stage Requirements 31
13.2.1 S-VB-D: Baseline Earth-Departure Stage 31
13.2.2 S-VB-A: Combined Ascent and Departure Variant 32
13.3 Solution A: Super Heavy with an Orbital Ascent Stage and Apollo Top 33

13.4 Solution B: SLS Ascent System with an Apollo Top 34
13.5 Selection Criteria for the Two Proposed Solutions 35

14 Source Treatment **36**

1 Purpose and Method

At least two mass states must be distinguished: total mass delivered to a specified low Earth parking orbit and assembled vehicle mass at the beginning of TLI. Apollo provides a useful historical reference because the S-IVB, CSM, and LM departed Earth as an integrated lunar stack. Modern Artemis architectures distribute major mission elements across separate launch and propulsion chains.

This study compares a single-launch integrated departure architecture with a separately launched crew-and-lander architecture supported by logistics flights. "Dual rocket" describes the two principal transportation chains, not necessarily two launches. Both architectures contain multiple spacecraft and propulsion systems. The comparison below treats mission completion, crew survival, and schedule reliability as separate outcomes.

The orbital comparison concerns Apollo's approximately two-hour low lunar orbit and the approximately 6.5-day near-rectilinear halo orbit (NRHO) used in the Artemis landing architecture studied here. HALO, the Gateway habitation module, is distinct from the halo orbit. Orbit period alone does not establish either rendezvous difficulty or emergency return time.

Landing stability is examined in Section 11 through vehicle dimensions, support geometry, touchdown dynamics, and a conditional probability model for tip-over on uneven terrain.

Configuration boundary. The ICPS/Orion numerical example remains an Artemis I reference configuration, not a complete future landing-mission specification. The lander comparisons refer to NRHO landing concepts, with and without Gateway as stated. NASA's March 2026 Inspector General report includes a February 2026 campaign revision; mission numbers and dated concepts should therefore not be read as a current flight manifest.[4]

2 Mass Accounting and Rocket Equation

Define m_D as dry stage/structural mass, m_U as propellant consumed, m_R as retained propellant, and m_L as delivered spacecraft/cargo. Then

$$m_0 = m_D + m_R + m_U + m_L, \qquad m_f = m_D + m_R + m_L.$$

The ideal rocket equation is

$$\Delta v = I_{\mathrm{sp}} g_0 \ln\left(\frac{m_0}{m_f}\right), \qquad g_0 = 9.806\,65\,\mathrm{m/s^2}.$$

Solving for useful delivered mass gives

$$m_L = \frac{m_U}{\exp\left(\Delta v/(I_{\mathrm{sp}} g_0)\right) - 1} - m_D - m_R.$$

Launch mass, parking-orbit mass, TLI ignition mass, and post-TLI mass are not interchangeable.

3 Numerical Vehicle Data

3.1 Apollo: S-IVB, CSM, and LM

Apollo 11 spacecraft masses used in this working comparison are:

Component	Mass (lb)	Mass (kg)
Command Module	12,250	5,557
Service Module	51,160	23,206
Lunar Module	33,200	15,059
Spacecraft–LM Adapter	4,045	1,835

For the LM, ascent-stage dry mass was about 4,804 lb, descent-stage dry mass 4,483 lb, RCS propellant 604 lb, descent propulsion propellant 18,100 lb, and ascent propulsion propellant 5,214 lb. Thus

$$m_{\mathrm{LM,dry}} \approx 4.21 \text{ t}, \qquad m_{\mathrm{LM,prop}} \approx 10.85 \text{ t}.$$

A working Saturn V 500-series S-IVB specification is:

Parameter	Working value
Empty mass	13.5 t
Propellant	109.5 t
Gross mass	123 t
Engine	1 J-2
Vacuum thrust	1,033.1 kN
Vacuum specific impulse	421 s
Propellants	LOX/LH_2

The AS-506 postflight evaluation gives the following actual complete-stack masses (including S-IVB, instrument unit, and spacecraft):[11]

$$M_{\text{Earth orbit}} = 135.102 \text{ t}, \qquad M_{\mathrm{TLI,ignition}} = 134.046 \text{ t},$$

$$M_{\mathrm{TLI,cutoff}} = 63.189 \text{ t}, \qquad M_{\text{TLI,end decay}} = 63.116 \text{ t}.$$

Ignition-to-end-decay mass loss is consequently 70.930 t. This is an assembly mass difference, including startup, shutdown, and auxiliary expenditures; it is not simply mainstage engine consumption. Table 20-9 reports 70.684 t of second-burn mainstage propellant. The earlier 134.046 t / 63.22 t pair and 74.58 t consumption are superseded: they mixed incompatible source values. The revised ledger below uses postflight event definitions consistently.

3.2 SLS, ICPS, Orion, and ESM

Published Artemis I reference values and separately identified configuration proxies are:

Parameter	Working value
Orion gross launch mass (published reference)	32.65 t
Orion mass at TLI (published reference)	24.05 t
Post-TLI Orion mass	Source units conflict; see ledger
Orion usable propellant	7.25 t
ESM dry mass (secondary-source working value)	6.185 t
ESM fueled/launch mass (configuration dependent)	15.461 t
ESM usable propellant (working value)	8.6 t

The ICPS is represented approximately by:

ICPS parameter	Working value
Dry mass	~ 3.8 t
Fully loaded mass	~ 32–33 t
Propellant	~ 28–29 t
Engine	1 RL10B-2
Vacuum thrust	~ 110 kN
Propellants	LOX/LH_2

A rough launch-configuration upper-stack mass is therefore of order

$$M_{\text{ICPS+adapter+Orion}} \sim 66 \text{ t},$$

depending on exact mission configuration. This is not the TLI ignition mass: ICPS propellant is consumed before TLI and Orion sheds ascent hardware. A burn-by-burn ledger is required.

The NASA reference guide gives 32.7 t fueled for ICPS and 0.8 t for the Orion stage adapter. Its post-TLI Orion entry pairs 51,500 lb with 23,850 kg; these are not equivalent (51,500 lb is approximately 23,360 kg). That entry is not used to infer a measured separation mass.[12]

3.3 Starship Human Landing System

Where a current authoritative HLS-specific dry mass is unavailable, this working study uses the commonly cited reference-Starship values as a proxy:

Parameter	Working value
Dry mass proxy	~ 100 t
Propellant capacity	$\sim 1,200$ t
Fully fueled mass proxy	$\sim 1,300$ t
Propellants	LOX/CH_4
HLS payload capability	100+ t class

The 100-t dry value is a secondary-source/reference-Starship proxy, not a measured HLS flight dry mass. More recent generic Starship specifications have quoted approximately 1,600 t of spacecraft propellant; that represents a different revision and should be treated as a separate sensitivity case.

For scale,

$$\frac{1,300}{134.046} \approx 9.70.$$

Thus the fully fueled 1,300-t proxy is about 9.7 times the mass of the entire Apollo 11 vehicle at TLI ignition. This does not establish efficiency; it shows why all tanker/refueling mass needed to establish that departure state must be counted.

3.4 Blue Moon Mark II

Secondary-source values are used where a complete current engineering mass breakdown is not public:

Parameter	Working value
Dry mass	~ 16 t
Launch/wet mass	> 45 t
Loaded-minus-dry mass	> 29 t
Height	~ 16 m
Main engines	3 BE-7
Propellants	LOX/LH_2

The > 29 t difference must not automatically be identified entirely as usable propellant because consumables and configuration-dependent hardware may be included. The associated Blue Origin Transporter architecture has been described as capable of moving approximately 100 t of propellant from Earth orbit toward lunar orbit.

4 Summary Mass Table

System	Dry/hardware	Propellant	Loaded/relevant mass	Status
Apollo S-IVB	13.5 t	109.5 t	123 t	Flown/reference
Apollo 11 LM	4.21 t	10.85 t	15.06 t	Flown
Apollo 11 complete TLI stack	—	—	134.046 t	Postflight
ICPS	~ 3.8 t	~ 28–29 t	~ 32–33 t	Flown
Orion spacecraft	—	7.25 t usable	24.05 t at TLI	Published reference
ESM	6.185 t	8.6 t working	15.461 t	Mixed/config.-dependent
Starship/HLS proxy	~ 100 t	$\sim 1,200$ t	$\sim 1,300$ t	Dev/proxy
Blue Moon Mk2	~ 16 t	> 29 t loaded-minus-dry	> 45 t	Dev/secondary

5 Architecture-Level Transport Efficiency

For architectures requiring multiple launches,

$$M_{\mathrm{LEO,total}} = \sum_{i=1}^{N} M_{\mathrm{LEO},i}.$$

Define the study-specific metric

$$\eta_{\mathrm{TLI}} = \frac{m_{\mathrm{useful,TLI}}}{M_{\mathrm{LEO,total}}}.$$

For orbital-refueling architectures, the denominator includes supporting propellant/tanker launches. This is not a standard universal rocket-performance parameter; it is defined here to expose architecture-level mass efficiency.

6 Normalized Versus Mission-Native Analysis

A normalized comparison places each departure system conceptually in the same reference parking orbit and assigns a common target departure condition. This isolates dry mass, propellant, specific impulse, thrust, payload, and burn duration.

A mission-native comparison instead preserves actual parking orbit, ascent expenditure, orbital assembly, refueling, spacecraft configuration, reserves, and departure sequence. The first asks how capable the propulsion stage is; the second asks how capable the complete transportation architecture is.

7 Apollo and Artemis as Different Mass Architectures

Apollo 11 began TLI with approximately 134.046 t of integrated vehicle mass, including the Earth-departure stage, CSM, LM, and structural hardware. Artemis I Orion itself was approximately 24.05 t at TLI. These are not equivalent system quantities because Orion's TLI mass does not contain the lunar lander.

For a landing architecture, a useful schematic accounting is

$$M_{\text{Artemis,total}} = M_{\text{Orion chain}} + M_{\text{lander chain}} + M_{\text{refueling/transport chain}},$$

with careful treatment of docking and propellant transfer to prevent double counting.

7.1 Vehicle Dynamics Beyond Mass Ratio

Mass ratio is a fundamental parameter in launch-vehicle analysis, but by itself it does not completely characterize the capability of a vehicle or upper stage. Two stages may have identical mass ratios while providing substantially different mission performance because of differences in specific impulse, thrust, burn duration, structural mass, staging, and trajectory. For lunar missions in particular, the relevant question is not simply how much propellant a vehicle carries, but how much useful mass the vehicle can move from one required orbital state to another.

A useful vehicle-dynamics analysis therefore begins with the rocket equation and extends it to include propulsion performance, finite-burn effects, orbital mechanics, staging, and mission-specific Δv requirements.

7.1.1 Mass Ratio and Propellant Fraction

For a single stage, the mass ratio is

$$MR = \frac{m_0}{m_f}, \tag{1}$$

where m_0 is the mass at ignition and m_f is the mass following the burn. The initial mass can be decomposed as

$$m_0 = m_{\text{dry}} + m_p + m_L, \tag{2}$$

where m_{dry} is the stage dry mass, m_p is the usable propellant mass, and m_L represents payload and any downstream hardware that must also be accelerated.

Following propellant depletion,

$$m_f = m_{\text{dry}} + m_L. \tag{3}$$

It is also useful to define the propellant mass fraction,

$$\lambda_p = \frac{m_p}{m_0}, \tag{4}$$

and the structural coefficient,

$$\epsilon_s = \frac{m_{\text{dry}}}{m_{\text{dry}} + m_p}. \tag{5}$$

The structural coefficient permits stages of substantially different physical size to be compared on the basis of the fraction of stage mass devoted to hardware rather than propellant.

7.1.2 Specific Impulse and Exhaust Velocity

Mass ratio becomes dynamically meaningful only when combined with propulsion performance. The effective exhaust velocity is

$$v_e = I_{sp} g_0, \tag{6}$$

where I_{sp} is the engine specific impulse and

$$g_0 \approx 9.80665 \text{ m/s}^2. \tag{7}$$

Consequently, two stages having the same mass ratio can have substantially different velocity capability if their propulsion systems have different specific impulses. This distinction is especially important when comparing high-specific-impulse cryogenic upper stages with lower-specific-impulse propulsion systems.

7.1.3 Ideal Delta-V

The ideal velocity increment available from a stage is given by the Tsiolkovsky rocket equation,

$$\Delta v_{\text{ideal}} = I_{sp} g_0 \ln\left(\frac{m_0}{m_f}\right). \tag{8}$$

Substituting the individual mass components gives

$$\Delta v_{\text{ideal}} = I_{sp} g_0 \ln\left(\frac{m_{\text{dry}} + m_p + m_L}{m_{\text{dry}} + m_L}\right). \tag{9}$$

For a multistage vehicle, the total ideal velocity increment is obtained by summing the contribution of each stage,

$$\Delta v_{\text{ideal,total}} = \sum_i I_{sp,i} g_0 \ln\left(\frac{m_{0,i}}{m_{f,i}}\right). \tag{10}$$

Care must be taken when applying this expression because the payload of one stage may include the complete mass of all subsequent stages and spacecraft.

7.1.4 Payload Sensitivity

The rocket equation can also be inverted to determine the payload that a stage can accelerate through a specified velocity increment. Let

$$R = \exp\left(\frac{\Delta v}{I_{sp} g_0}\right). \tag{11}$$

Then

$$R = \frac{m_{\text{dry}} + m_p + m_L}{m_{\text{dry}} + m_L}. \tag{12}$$

Solving for payload gives

$$m_L = \frac{m_p}{R - 1} - m_{\text{dry}}, \tag{13}$$

or equivalently,

$$\boxed{m_L = \frac{m_p}{\exp\left(\Delta v/(I_{sp} g_0)\right) - 1} - m_{\text{dry}}}. \tag{14}$$

This relationship is particularly useful for comparing lunar architectures. Rather than asking only how massive an upper stage is, one can ask how much useful spacecraft mass the stage can inject onto a specified trajectory.

7.1.5 Thrust-to-Weight Ratio

The rocket equation does not contain thrust explicitly. Nevertheless, thrust has an important effect on actual vehicle performance. The instantaneous thrust-to-weight ratio is

$$\text{TWR} = \frac{T}{mg}, \tag{15}$$

where T is total engine thrust and m is instantaneous vehicle mass.

During ascent from the Earth's surface, thrust-to-weight ratio strongly affects gravity losses. For an upper stage already in orbit, the effect is different because the vehicle is already in free fall. Nevertheless, thrust determines burn duration, and a sufficiently long burn can no longer be treated as an instantaneous or impulsive orbital maneuver.

7.1.6 Propellant Flow Rate and Burn Duration

For approximately constant thrust and specific impulse, propellant mass flow is

$$\dot{m} = \frac{T}{I_{sp} g_0}. \tag{16}$$

An approximate burn duration is therefore

$$t_b = \frac{m_p}{\dot{m}}. \tag{17}$$

Burn duration is particularly important for translunar injection. An ideal orbital maneuver is often modeled as an instantaneous change in velocity. A real TLI maneuver instead occurs over a finite arc of the parking orbit.

A high-thrust upper stage can concentrate a larger fraction of its energy addition near the desired point in the orbit. A lower-thrust stage may require a longer burn, requiring numerical trajectory integration rather than a purely impulsive approximation.

7.1.7 Ideal Delta-V Versus Effective Delta-V

The rocket equation describes the ideal velocity capability of the propulsion system. Actual trajectory performance can be represented schematically as

$$\Delta v_{\text{effective}} = \Delta v_{\text{ideal}} - \Delta v_{\text{gravity}} - \Delta v_{\text{drag}} - \Delta v_{\text{steering}} - \Delta v_{\text{other}}. \tag{18}$$

For launch from the Earth's surface, atmospheric drag and gravity losses can be significant. For an upper stage performing TLI from an established parking orbit, atmospheric drag is effectively absent, but finite-burn, steering, guidance, and other trajectory effects remain.

Thus, quoted ideal Δv should not automatically be interpreted as mission-usable Δv.

7.1.8 Orbital Energy

Velocity alone is not sufficient to characterize a lunar trajectory because the significance of a particular velocity depends upon position within the gravitational field. A more useful quantity is specific orbital mechanical energy,

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r}, \tag{19}$$

where μ is the gravitational parameter of the central body and r is distance from its center.

For a Keplerian orbit,

$$\mathcal{E} = -\frac{\mu}{2a}, \tag{20}$$

where a is the semimajor axis. Combining these relations produces the vis-viva equation,

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right). \tag{21}$$

This provides the connection between propulsion performance and trajectory. A propulsion system supplies Δv, while the resulting change in velocity changes the spacecraft's orbital energy and therefore its orbit.

7.1.9 Translunar Injection

For a simplified patched-conic treatment, a spacecraft initially in a circular Earth parking orbit has velocity

$$v_{\text{LEO}} = \sqrt{\frac{\mu_E}{r_{\text{LEO}}}}. \tag{22}$$

If the translunar trajectory is approximated as an elliptical transfer orbit having perigee at the parking orbit and apogee near lunar distance, then

$$a_{\text{TLI}} = \frac{r_{\text{LEO}} + r_{\text{Moon}}}{2}. \tag{23}$$

The required velocity at transfer-orbit perigee is

$$v_{\mathrm{TLI}} = \sqrt{\mu_E \left(\frac{2}{r_{\mathrm{LEO}}} - \frac{1}{a_{\mathrm{TLI}}} \right)}. \tag{24}$$

The corresponding ideal impulsive TLI requirement is therefore

$$\Delta v_{\mathrm{TLI}} = v_{\mathrm{TLI}} - v_{\mathrm{LEO}}. \tag{25}$$

This quantity can be compared directly with the available Δv of an upper stage carrying a specified spacecraft stack.

A more accurate lunar trajectory calculation must include the Moon's motion, three-body effects, departure geometry, launch date, inclination, targeting requirements, and finite burn duration. Nevertheless, the patched-conic calculation provides a useful first-order comparison between candidate vehicles.

7.1.10 Delta-V Margin

For each major mission maneuver, the relevant engineering quantity is the difference between available and required velocity capability,

$$\Delta v_{\mathrm{margin}} = \Delta v_{\mathrm{available}} - \Delta v_{\mathrm{required}}. \tag{26}$$

A positive ideal margin is necessary but is not, by itself, sufficient to establish mission feasibility. The margin must also accommodate trajectory corrections, guidance errors, dispersions, propellant reserves, residual propellant, and other operational contingencies.

Consequently, a configuration that satisfies the nominal maneuver by only a very small theoretical margin should not be considered equivalent to one having substantial operational reserve.

7.1.11 Staging and Discarded Mass

A complete vehicle analysis must also account for when hardware is discarded. The importance of a kilogram of mass depends strongly upon how far through the mission it must be transported.

For example, a kilogram discarded before TLI does not need to be accelerated through TLI. A kilogram retained through TLI but discarded before lunar descent does not need to be landed. Conversely, a kilogram that travels to the lunar surface and subsequently returns must potentially be accelerated through several major maneuvers.

For this reason, the mass model should be evaluated immediately before and after every major staging event:

$$m_i^- \longrightarrow m_i^+, \tag{27}$$

where m_i^- denotes vehicle mass immediately before separation and m_i^+ denotes the remaining vehicle mass immediately afterward.

This staging analysis is especially important when comparing architectures whose total launch masses are very different but whose masses delivered to the lunar vicinity or lunar surface may be much closer.

7.1.12 Usable Propellant, Reserves, and Residuals

Nominal tank capacity should not automatically be treated as available propellant. It is useful to distinguish

$$m_{p,\text{loaded}} = m_{p,\text{usable}} + m_{p,\text{reserve}} + m_{p,\text{residual}}. \tag{28}$$

Propellant may be unavailable for nominal mission maneuvers because of engine shutdown requirements, trapped residuals, ullage requirements, guidance reserves, boiloff, thermal conditioning, or mission contingency requirements.

For cryogenic stages, elapsed time can additionally make propellant mass a time-dependent quantity,

$$m_p = m_p(t), \tag{29}$$

which becomes increasingly important for architectures requiring long loiter periods, propellant transfer, or multiple rendezvous operations.

7.1.13 Mission-State Mass and Energy Ledger

A useful method for comparing lunar architectures is therefore to construct a mass and energy ledger at each major mission state. A generic ledger has the form

Mission state	Mass	Propellant	Velocity	Specific energy	Required Δv	Remaining Δv
LEO insertion	m_1	$m_{p,1}$	v_1	$\mathcal{E}_1$	—	Δv_1
Before TLI	m_2	$m_{p,2}$	v_2	$\mathcal{E}_2$	Δv_{TLI}	Δv_2
After TLI	m_3	$m_{p,3}$	v_3	$\mathcal{E}_3$	—	Δv_3
Lunar arrival	m_4	$m_{p,4}$	v_4	$\mathcal{E}_4$	Capture/LOI	Δv_4
Before descent	m_5	$m_{p,5}$	v_5	$\mathcal{E}_5$	Descent	Δv_5
Lunar surface	m_6	$m_{p,6}$	—	—	Ascent	Δv_6
Before Earth return	m_7	$m_{p,7}$	v_7	$\mathcal{E}_7$	TEI	Δv_7

Such a ledger makes explicit what mass is present, what propellant remains, what orbital state has been achieved, and what propulsion capability remains at every important point in the mission.

7.1.14 Assembly-Specific Mission Ledgers

Read the evidence labels before comparing numbers. R means a reported postflight value; **C** means a calculation from reported values; **S** means a deliberately specified scenario, *not* a measured or manufacturer-confirmed vehicle state. Apollo has a postflight reconstruction. For the other systems, complete public flight/configuration inventories were not established by the sources consulted; the numerical examples below fill the accounting model, not that evidence gap. The scenario inputs are disclosed so that they can be replaced when better data become available. None is a certified mission prediction.

Here m is total mass of the assembly named in the row. P is the propellant inventory covered by that table, including any specified reserve; it is not all usable maneuver propellant. For Apollo, P includes *main-system liquid propellant only*, excluding separate RCS inventories, ullage gases, pressurant, and life-support/fuel-cell reactants. For the scenarios, P is the modeled propulsion inventory. Values are rounded, and mass closure is performed before rounding. A positive final inventory alone does not demonstrate feasibility.

At each burn the scenario propagation is

$$m_{i+1} = m_i \exp\left[-\frac{\Delta v_i}{I_{sp,i} g_0}\right], \qquad P_{i+1} = m_{i+1} - B_i,$$

where B_i is all retained non-propellant mass. Transfers add to the receiving vehicle and subtract from the supplying vehicle; staging subtracts discarded mass. No number in these models is obtained by assuming that every kilogram above bare dry mass must be usable propellant.

Common maneuver convention. The fast-transfer reference uses 3.20 km/s for Earth departure, 0.45 km/s for NRHO arrival, and 2.75 km/s each for descent and ascent. These are rounded architecture-study budgets from NASA's October 2019 orbit comparison, not a selected HLS flight trajectory.[15] Slower ballistic transfers, different departure orbits, and optimized burns can give different results. The lander budgets are applied directly as total modeled propulsive increments; no second gravity-loss allowance is added. Separate RCS usage, boiloff, transfer losses, consumable depletion, cargo unloading, and docking impulses are zero in the base cases. This simplifying assumption makes the examples optimistic, not complete operations budgets. Velocity vectors and specific energy are not inferred from mass alone; those columns require a dated trajectory and a stated reference frame.

Apollo 11: postflight mass ledger.

State / assembly	m (t)	P (t)	Basis / accounting
Earth orbit, first-burn end decay	135.102	104.746	R mass; C main inventory = 75.614 t S-IVB + 29.132 t spacecraft.
TLI ignition: complete stack	134.046	103.584	R mass; C main inventory = 74.452 t S-IVB + spacecraft.
TLI cutoff: complete stack	63.189	32.590	R mass; S-IVB liquid inventory 3.458 t.
TLI end decay: complete stack	63.116	32.517	R mass; S-IVB liquid inventory 3.385 t.
Transposition/docking: CSM + LM	43.893	29.132	R mass; C sum of SPS, descent, and ascent loads.
Lunar capture ignition: CSM + LM	43.573	∼28.94	R mass; C approximate main inventory after preceding SPS burns.
Lunar capture cutoff: CSM + LM	32.676	∼18.05	R mass; C approximate main inventory.
Circularization cutoff: CSM + LM	32.162	∼17.54	R mass; C approximate main inventory before descent separation.

Masses come from AS-506 Tables 20-5 and 20-7 for the launch stack, and Apollo 11 Mission Report Table A-I for the spacecraft.[11, 1] S-IVB liquid totals sum the in-tank and below-tank LOX/LH_2 entries; not all remaining liquid is usable. Small report-to-report spacecraft mass differences are preserved rather than silently merged. The approximate SPS reconstructions subtract burn endpoint mass differences and neglect concurrent minor losses.

Apollo 11: lunar surface and return states.

State / assembly	m (t)	P (t)	Basis / accounting
CSM + LM just before separation	32.096	∼17.54	R mass; C main inventory, neglecting minor changes since circularization.
LM at touchdown	7.327	2.725	R mass; C 0.349 t descent + 2.376 t ascent inventory; excludes RCS.
LM ascent stage at liftoff	4.888	2.376	R mass; C ascent load; descent stage remains on the surface.
LM ascent stage at jettison	2.478	0.182	R mass and main-system remainder; separate RCS remains too.
CSM at TEI ignition	16.767	∼6.92	R mass; C approximate remaining SPS inventory.
CSM at TEI cutoff	12.153	∼2.30	R mass; C approximate remaining SPS inventory.
CSM before service-module separation	12.091	2.297	R mass and SPS remainder; latter leaves with service module.
CM at entry interface	5.486	0.000	R mass; no main-engine inventory aboard CM. CM RCS is excluded, not absent.

Apollo 11 Mission Report, Table A-I and Sections 8.11 and 9.13.[1] Separate propulsion systems must remain separate in a reserve analysis: the 402 lb ascent-system remainder is not the sum of all remaining LM propellant. The report also lists 315 lb of LM RCS propellant at jettison.

ICPS/Orion assumptions. The following **S** case uses published reference masses of 32.7 t fueled ICPS, 0.8 t Orion stage adapter, and 24.05 t Orion.[12] It assumes ICPS retained inert mass 3.8 t, initial main propellant 28.9 t, constant thrust 110 kN, and $I_{sp} = 462$ s. The published 22 s perigee-raise and approximately 18 min TLI timeline supplies reference durations.[13] These are not reconstructed flight tank readings. Applying loaded ICPS mass at orbital insertion neglects pre-burn venting/conditioning; smaller payloads and unlisted adapter contents are excluded. The stage mass assumption absorbs non-main-system inventories without treating them as expendable main propellant.

ICPS/Orion: numerical reference case, all rows S.

State / assembly	m (t)	P (t)	Basis / accounting
Earth orbit before perigee raise	57.55	28.90	ICPS + OSA + Orion; P is ICPS main propellant only.
TLI ignition	57.02	28.37	Perigee raise consumes 0.53 t in model.
TLI cutoff before separation	30.79	2.14	TLI consumes 26.22 t; model ideal increment about 2.79 km/s.
Orion after separation	24.05	7.25	P now denotes assumed Orion maneuver propellant; not the ICPS inventory.
Separated ICPS + OSA	6.74	2.14	Includes remaining ICPS propellant for later operations; not part of Orion.

This is an Artemis I-like departure example from its elliptical staging orbit, not the normalized 3.20 km/s circular-LEO case. NASA's reference-guide post-TLI Orion units conflict; the model retains 24.05 t across separation and assumes no Orion expenditure there, rather than inventing a

corrected observed value. Artemis I flew DRO, not NRHO. Its later trajectory is not spliced into the following NRHO scenario.

Orion: separate NRHO reference case, all rows S.

State / assembly	m (t)	P (t)	Basis / accounting
Before lunar arrival maneuvers	24.05	7.25	24.05 t departure reference; B = 16.80 t and assumed Isp = 316 s.
After 0.45 km/s NRHO arrival	20.80	4.00	Consumes 3.25 t; no separate correction or RCS allowance.
Docking / surface interval	20.80	4.00	Orion alone; add lander mass only while physically docked.
After 0.45 km/s Earth departure	17.99	1.19	Consumes 2.81 t; later corrections omitted.
Crew-module landing reference	8.25	N/A	Published landing-mass reference, not propagated entry mass; ESM discarded.

The 8.25 t endpoint is a published reference, not an as-flown result.[12] There is no continuous mass reconstruction across service-module disposal and entry ablation in this simplified model. As an independent observed check, NASA reported about 12,100 lb (5.49 t) of propellant used on Artemis I by flight day 24; that DRO-mission consumption is not substituted for the NRHO scenario.[14]

Starship HLS assumptions. This **S** case retains the draft's 100 t dry / 1,200 t propellant proxy, adds an assumed 10 t retained payload, and assumes a constant effective $I_{sp} = 380$ s for every burn. The same performance for all engines and phases is an optimistic simplification. It is not the current HLS design mass or a verified landing-engine specification. Earth-orbit arrival inventory is set to 20 t solely to show the refueling transaction; departure results are independent of that arbitrary arrival inventory once the prescribed fill is achieved.

Starship HLS: numerical proxy case, all rows S.

State / assembly	m (t)	P (t)	Basis / accounting
LEO arrival before refueling	130.0	20.0	Assumed dry + payload = 110 t; initial propellant is a scenario choice.
Refueled / TLI ignition	1310.0	1200.0	Receives 1,180 t net propellant; gross tanker delivery must also cover losses.
TLI cutoff	555.1	445.1	3.20 km/s modeled increment.
NRHO arrival / crew docking	491.9	381.9	Additional 0.45 km/s; loiter losses omitted.
Lunar surface	235.2	125.2	2.75 km/s descent; all payload retained.
Return to NRHO	112.4	2.4	2.75 km/s ascent; final inventory is not a validated reserve.

This proxy leaves only 2.44 t. If the assumed terminal reserve is 5 t, the required initial propellant becomes 1229.8 t; the 1,200 t case misses that illustrative reserve by 29.8 t at departure. That is a result of this proxy and budget, not evidence that the real HLS is infeasible. A different tank capacity, dry mass, payload, engine performance, or transfer strategy changes the answer.

Blue Moon Mark II assumptions. This **S** case uses the draft's 16 t dry proxy, an assumed 2 t retained payload, $I_{sp} = 450$ s, and 1 t of propellant remaining on return to NRHO. The required NRHO mass is solved backward, rather than declaring the draft's "greater than 45 t" launch mass to be a tank-capacity specification:

$$m_{\text{NRHO,full}} = (16 + 2 + 1) \exp\left[\frac{5,500}{450 g_0}\right] = 66.073 \text{ t}.$$

The departure case assumes capacity for the resulting 48.073 t inventory and refilling to the same state in NRHO. Neither that capacity nor that loading is verified for Blue Moon. LEO arrival is set to 45 t (27 t propellant) only as a round-number illustration, not a measured launch or insertion mass.

Blue Moon Mark II: back-solved reference case, all rows S.

State / assembly	m (t)	P (t)	Basis / accounting
LEO before modeled refueling	45.00	27.00	Assumed insertion state; excludes any attached transporter.
TLI ignition	66.07	48.07	Add 21.07 t in Earth orbit; assumed tank capacity required.
TLI cutoff	32.00	14.00	3.20 km/s modeled increment.
NRHO arrival before refueling	28.89	10.89	Additional 0.45 km/s; losses omitted.
NRHO refueled / crew docking	66.07	48.07	Receives 37.18 t from transporter.
Lunar surface	35.43	17.43	2.75 km/s descent.
Return to NRHO	19.00	1.00	2.75 km/s ascent; terminal reserve specified by construction.

A 45 t vehicle with 18 t retained non-propellant mass provides only about 4.04 km/s before exhausting its modeled propellant, below the 5.50 km/s surface round-trip budget. That comparison shows why launch mass, lunar-orbit refueled mass, and tank capacity must not be conflated. It does not establish the actual Blue Moon mass or maximum propellant load.

Supporting transporter assumptions. No validated transporter dry mass or full tank inventory was established. The next table is therefore a **synthetic 100 t delivery example**, not a Blue Origin vehicle estimate. Assume 20 t dry mass, $I_{sp} = 450$ s, 100 t cargo propellant at lunar arrival, and 2 t own-use propellant retained afterward. A fast-transfer departure is back-solved as

$$m_{T,0} = (20 + 100 + 2) \exp\left[\frac{3,650}{450 g_0}\right] = 278.976 \text{ t}.$$

A zero-loss transfer illustrates ledger closure. Propellant cargo stays aboard during the modeled burns and is not consumed by the transporter. This excludes the real concept's possible intermediate orbit raising and refueling sequence.

Transporter: synthetic accounting example, all rows S.

State / assembly	m (t)	P (t)	Basis / accounting
Initial orbital assembly	22.00	2.00	Chosen baseline: dry structure + retained own-use reserve; cargo empty.
Refueled / departure	278.98	258.98	P = 158.98 t own-use + 100 t delivery cargo.
After 3.20 km/s TLI	135.10	115.10	Own-use propellant 15.10 t; cargo still 100 t.
After 0.45 km/s NRHO insertion	122.00	102.00	2 t own-use reserve + 100 t delivery cargo.
After Blue Moon transfer	84.82	64.82	Delivers 37.18 t; cargo remainder 62.82 t; own-use reserve 2 t.

The required departure mass is calculated, not a published transporter specification. No claim is made that this mass fits a particular launcher, tank, or orbital assembly plan. Reuse or Earth return would require an additional reserve and trajectory budget.

Sensitivity and evidence limits. Model uncertainty is structural, not just rounding. The following are alternative inputs, not statistical confidence intervals or manufacturer tolerances.

HLS sensitivity: required initial propellant (S).

Case	Initial P (t)	End P (t)	Interpretation
HLS: dry 100 t, Isp 360 s	1426	5	Required departure P for 10 t payload and specified terminal reserve.
HLS: dry 100 t, Isp 380 s	1230	5	Required departure P for 10 t payload and specified terminal reserve.
HLS: dry 120 t, Isp 360 s	1673	5	Required departure P for 10 t payload and specified terminal reserve.
HLS: dry 120 t, Isp 380 s	1443	5	Required departure P for 10 t payload and specified terminal reserve.
HLS: dry 150 t, Isp 360 s	2043	5	Required departure P for 10 t payload and specified terminal reserve.
HLS: dry 150 t, Isp 380 s	1762	5	Required departure P for 10 t payload and specified terminal reserve.

For Blue Moon, with retained non-propellant mass varied from 16 to 22 t and effective specific impulse from 430 to 460 s, the same surface-round-trip and 1 t terminal reserve require approximately 57.5–84.8 t at NRHO departure. For the transporter, varying dry mass from 10 to 30 t at 450 s changes the 100 t delivery departure mass from about 256.1 to 301.8 t. These ranges express the selected parameter sweeps only; actual hardware may fall outside them.

Count each Earth-to-orbit delivery once. Transferred propellant is an internal architecture flow, not an extra launch contribution. Tankers staying in Earth orbit do not become part of the lander's TLI assembly. The remaining evidence gap is the actual vehicle-specific input set: measured HLS/Blue Moon dry mass, tank capacity, payload, arrival inventory, engine performance by phase, losses, and reserves. The tables now contain numerical reference cases, but only the Apollo table is a postflight reconstruction. They cannot support a definitive ranking of flight efficiency or crew reliability.

7.1.15 Implications for Vehicle Comparison

The resulting comparison is therefore not simply one of gross vehicle mass or propellant capacity. A useful lunar-vehicle comparison should include at least

$$\{m_{\text{dry}}, m_p, m_L, I_{sp}, T, t_b, \Delta v, \mathcal{E}, \epsilon_s\}, \tag{30}$$

together with the sequence of staging events and the Δv requirements of the selected trajectory.

This distinction is important when comparing vehicles such as the S-IVB, ICPS, EUS, Apollo spacecraft, and proposed lunar lander systems. A larger vehicle does not necessarily deliver proportionally more useful lunar payload, while a vehicle with a favorable mass ratio does not necessarily have sufficient propulsion or trajectory capability to perform the required mission.

The more physically meaningful question is therefore not simply

> How much does the vehicle weigh?

but rather

> Given its dry mass, propellant, propulsion system, staging sequence, and trajectory, how much useful spacecraft mass can the vehicle move from one required orbital state to the next while maintaining adequate mission margin?

This mass–energy–trajectory formulation provides a common basis on which historical Apollo hardware, current SLS-derived systems, and proposed lunar architectures can be compared.

8 Single-Launch and Separate-Launch Architecture Complexity

Apollo placed its crew spacecraft and lander on one Saturn V and one integrated TLI stack. That did not eliminate docking: CSM transposition, docking, and LM extraction followed TLI, and another docking followed lunar ascent. The useful contrast is integrated deployment versus independent deployment, rather than one spacecraft versus two.

Dependency	Apollo integrated launch	Separate crew / lander chains
Launch campaign	One Saturn V for the landing mission.	Crew launch, lander launch, and architecture-dependent supporting launches.
Earth-orbit logistics	No orbital propellant transfer for the lunar mission.	Cryogenic storage, rendezvous, transfer, and verification add interfaces.
Lunar arrival	CSM and LM travel together after extraction.	Independent targeting, arrival, readiness, and lifetime constraints must coincide.
Crew-transfer docking	Post-TLI extraction docking and post-ascent docking.	Pre-descent and post-ascent crew-transfer docking, plus separate logistics dockings; Gateway changes the sequence.
Schedule recovery	Launch failure or major stack defect affects the complete mission.	An uncrewed logistics failure may be retried before committing crew, if time and resources permit.
Backup accessibility	LM reachable through docking tunnel during much of outbound flight.	Lander resources inaccessible to Orion crew until rendezvous and successful transfer.
Shared failure exposure	Co-launched vehicles share launch and nearby accident exposure.	Independent launches separate some hazards but add software, operations, and interface dependencies.

More required events create more ways to delay or lose the landing objective. They do not, by themselves, establish a greater probability of crew death. Predeployment can move hazardous logistics operations ahead of crew exposure; that benefit depends on effective checkout and the ability to defer crew launch. An unsuccessful uncrewed tanker flight is not equivalent to a failed crewed ascent. Conversely, successful predeployment does not create a lifeboat next to Orion during Earth–Moon transit.

8.1 A Reliability Model with Separate Outcomes

Let A denote all predeployment readiness conditions being satisfied before crew commitment, and let C denote crewed mission completion. Then

$$P(\text{mission completion}) = P(A)P(C \mid A).$$

A simplified independent, non-retryable series of operations has reliability $\prod_i p_i$. For illustration only, ten required independent operations each having probability 0.99 yield $0.99^{10} \approx 0.904$, rather than 0.99 for one. These are hypothetical values, not Apollo or Artemis reliability estimates. Retries, common hardware, weather, schedule limits, and correlated failures invalidate a naive application of this product to launch counts.

For mutually exclusive initiating-failure categories F_k (including a category for combinations), a phase-dependent crew-risk model is

$$P(\text{loss of crew}) = \sum_k P(F_k)P(\text{loss of crew} \mid F_k, \text{phase, available resources}).$$

A backup primarily changes the conditional consequence of a failure. Demonstrated Apollo recovery capability therefore supports a claim about failure tolerance; it does not supply a statistical failure

rate for a modern spacecraft. A numerical comparison requires consistent exposure intervals, failure trees, common-cause analysis, and tested recovery coverage for both architectures.

9 Emergency Resources, Reliability, and Crew Survivability

9.1 Apollo's Attached Lunar Module as an Emergency Resource

Apollo 13 demonstrated the value of carrying a second crew compartment with oxygen, batteries, water, carbon-dioxide removal, and independent propulsion. After the service-module accident, the crew used the LM as a lifeboat and its descent engine for return-trajectory maneuvers. The spacecraft's landing resources could be reassigned because the landing was abandoned.[2]

These were *backup functions of mission resources*, not a separate reserve fuel tank added solely for rescue. LM propellant remained in the LM and powered its own engine; it was not transferred into the service module. The emergency capability therefore depended on an operable engine, controls, structure, docking connection, and a feasible trajectory for the combined mass.

Recovery was constrained: water and electrical power had to be conserved, and command-module carbon-dioxide cartridges required an improvised adapter for LM use. The command module still supplied Earth-entry protection.[2] The LM was consequently an important but incomplete substitute for a failed CSM.

Mission phase	Apollo backup availability	Separate-launch Artemis comparison
Launch / before LM extraction	Shared launcher; no immediately accessible docked LM cabin.	Crew launch abort and onboard spacecraft provisions apply; distant lander is unavailable.
Outbound transit after extraction	Docked LM life support and unspent descent propulsion can support appropriate contingencies.	Prepositioned HLS cannot immediately support Orion; recovery depends on onboard systems and reachable trajectories.
Docked before descent	Both cabins accessible; propulsion help requires a compatible burn and adequate resources.	A docked HLS or operational Gateway may offer refuge; certify capacity, interfaces, and actual recovery maneuvers.
Crew separated / surface stay	Orbital CSM cannot replace a failed LM surface-ascent system.	Orion likewise cannot collect crew directly from the surface; spare lander cannot be assumed.
After ascent / before lander disposal	LM descent stage and its supplies are no longer present.	Remaining HLS supplies and connectivity depend on configuration and reserve policy.
After lander separation / Earth return	CSM relies on its remaining systems; Apollo 13-style full LM backup is gone.	Orion relies on remaining onboard systems after separation from lunar assets.

9.2 Chemo-Mechanical Backup: Storable Hypergolic Propulsion

The LM's emergency propulsion value also depended on how its engine could be started and fed. Hypergolic fuel and oxidizer ignite on contact under their specified operating conditions. The Apollo LM ascent and descent engines were pressure-fed: pressurant forced the liquids into the

combustion chamber without a turbopump. The manufacturer's reference identifies storability and closely spaced starts among the reasons for selecting these propellants.[8]

In this study, *chemo-mechanical backup* means the recovery advantage of an available propulsion system whose ignition comes from propellant chemistry and whose feed comes from pressure. It is an architectural description, not a claim that hypergolicity is itself a second engine or a recognized numerical reliability metric. Two distinct properties matter:

1. **Chemical ignition.** Removing the need for a separate chamber igniter removes that particular start dependency. It does not remove the need for correctly timed fuel and oxidizer flow, suitable inlet conditions, or a sound combustion chamber.

2. **Pressure-fed delivery.** Removing the turbopump removes its drive and startup dependencies. Feed pressure must still be maintained through intact tanks, regulators, lines, and valves. Hypergolic chemistry and pressure feeding are separate design choices; neither implies the other.

For Apollo's docked LM, these properties helped make unused landing propellant an actionable emergency resource. The independent spacecraft and its remaining power, controls, and engine supplied the backup; propellant chemistry reduced some of the dependencies needed to use it. This interpretation is consistent with the Apollo 13 recovery, rather than proof that any hypergolic system will survive the same initiating failure.[2, 8]

Storability and emergency readiness. Storable liquids avoid the deep-cryogenic propellant storage and conditioning requirements of liquid oxygen/hydrogen or liquid oxygen/methane. They still require temperature control within operating limits; "storable" does not mean immune to freezing, leakage, or degradation. For an emergency burn after a long coast, the relevant comparison is the entire readiness chain: propellant state, feed pressure, valve operation, ignition, attitude control, and navigation. Cryogenic propulsion can be engineered for repeated starts and long-duration storage; its demonstrated readiness must be evaluated for the intended mission.

Does this mean Apollo can carry more propellant? No general capacity ranking follows from hypergolicity. Propellant mass for a fixed tank volume depends on density, $m_p = \rho_p V_p$, while velocity capability depends on specific impulse and the complete vehicle mass ratio. Liquid hydrogen has low density and requires large tanks, despite its high propulsion performance.[10] For a bipropellant vehicle, compare the combined fuel and oxidizer tank volumes at the specified mixture ratio, including insulation, pressurization hardware, and usable residuals. A dense propellant may save tank volume without delivering more Δv per kilogram.

Apollo's contingency advantage was *availability*: unspent LM propellant could power an attached independent engine after the landing was abandoned. It was not proof of a greater total propellant capacity than the larger Artemis vehicles. Likewise, a tanker or lander carrying more propellant elsewhere does not increase the emergency maneuver capability of an isolated crew spacecraft.

Limits of the backup. Hypergolic propulsion is not electricity-free, an unlimited-restart guarantee, or a remedy for a failed feed valve, lost pressurant, tank rupture, or a damaged nozzle. Nor is it a source of breathable oxygen or cabin electrical power. Fuel and oxidizer are hazardous chemicals, so leakage and unwanted contact create their own risks. A nonfunctional ascent engine cannot be rescued merely by having unburned hypergolic propellant aboard. Separate batteries and controls remain essential to the useful independence of a backup propulsion system.

Apply the same credit to Orion. Orion's European Service Module also uses storable MMH/MON propellants and helium pressurization; this benefit is not exclusive to Apollo.[9] The distinction is that Apollo carried an additional LM propulsion and habitat system during outbound transit. Orion's own propulsion redundancy does not make a separately deployed HLS available before docking. A fair comparison therefore credits Orion's storable propulsion while separately assessing the cryogenic lander's readiness and the physical availability of a second system.

9.3 What Can and Cannot Be Credited to Artemis

Orion has its own subsystem redundancy; absence of an attached LM is not absence of all backup capability. Published Orion trajectory studies examine auxiliary thruster return maneuvers after main-engine failure. These cover specific failure and trajectory cases, not arbitrary loss of service-module resources.[6]

For this comparison, credit a second habitat only when crew can reach it before their current supplies or vehicle fail. Credit backup propulsion only when its engine, power, propellant, guidance, and structural load path remain usable. A distant fueled lander is neither an immediate life-support reserve nor a second propulsion system for an isolated Orion. A docked vehicle is a potential refuge, but its presence alone does not prove that it can propel the whole stack onto an Earth-return trajectory. Landing propellant consumed before an emergency cannot also be counted as unused return reserve.

NASA's March 2026 Inspector General report states that early crewed Artemis missions lacked a required in-space crew-rescue capability and identifies stranding following lander disablement or inability to dock as a critical consequence.[4] Apollo also had no standby lunar-surface rescue lander. Neither architecture should receive credit for an unequipped or unvalidated rescue mission.

9.4 Emergency Endurance and Recoverability

A first screening measure is the time until the first essential resource is exhausted in the degraded configuration:

$$t_{\text{endurance}} = \min_{j} \frac{Q_{j,\text{remaining}}}{\dot{Q}_{j,\text{emergency}}}.$$

Here j includes oxygen, water, carbon-dioxide absorption capacity, and usable stored energy. This constant-rate approximation must be replaced by time-dependent thermal and power analysis where appropriate. Habitat volume alone is not an endurance measurement. The usable recovery margin is

$$t_{\text{margin}} = t_{\text{endurance}} - \left(t_{\text{activation}} + t_{\text{wait}} + t_{\text{transfer}} + t_{\text{rendezvous/docking}}\right).$$

The destination must itself remain habitable and have an onward recovery path. A positive time margin is necessary, but propulsion, navigation, and docking constraints must also close. This evaluates the particular emergency rather than declaring one entire architecture universally safer.

10 Crew Safety: Two-Hour Low Lunar Orbit Versus 6.5-Day NRHO

10.1 Period Is Not Rescue Time

For a representative 100 km circular lunar orbit, using lunar radius 1,737.4 km and gravitational parameter $4,902.8\ \mathrm{km}^3/\mathrm{s}^2$,

$$T = 2\pi\sqrt{\frac{(1,737.4+100)^3}{4,902.8}} \approx 118\ \text{min}.$$

This two-body calculation explains the approximately two-hour Apollo timescale. It is not a model of NRHO, which depends on Earth–Moon three-body dynamics. The ratio $6.5 \times 24/2 \approx 78$ compares periods only; it is neither a risk multiplier nor a ratio of rescue times.

Apollo 11 lifted off at mission time 124:22:00.8 and docked at 128:03:00, a 3 h 41 min rendezvous sequence. Its TEI-to-splashdown interval was about 59 h 55 min. These are flown nominal durations, not universal abort limits.[1] A two-hour orbit does not guarantee a launch opportunity, rendezvous, or safe docking every two hours: plane alignment, relative phase, ascent capability, and reserves all matter. Apollo's low-inclination landing geometry also cannot simply be transplanted to a south-pole mission.

10.2 Published NRHO Abort Durations and Their Limits

A 2022 mission-design study reports baseline NRHO surface-to-staging-orbit abort transfers of approximately 0.5–2.5 days, *excluding rendezvous, proximity operations, and docking.* It gives about 5–8 days from Orion to Earth and roughly 7–15 days for combined abort timelines including staging-orbit stay. The studied Earth-return opportunity occupied about 1–2 days per NRHO revolution; missing it could impose about 4.5 days of additional waiting with supporting logistics.[5]

Those figures describe that study's mission assumptions, not certified limits for every HLS design or a prediction for a particular future flight. The table below deliberately labels flown Apollo durations and modeled NRHO values separately. Both require new trajectory solutions for a like-for-like emergency at a specified landing site, epoch, and degraded vehicle condition.

Safety consideration	Apollo-type low lunar orbit	NRHO landing architecture
Orbital period	About 2 hours.	About 6.5 days; not a mandatory wait before every abort.
Surface to accessible refuge	Apollo 11 nominal ascent-to-docking: 3 h 41 min.	2022 study: 12–60 h to staging orbit, plus rendezvous/docking; phase dependent.
Return to Earth	Apollo 11 nominal TEI-to-splashdown: about 60 h.	2022 study: about 5–8 days from Orion; departure-window waits may add time.
Time-critical cabin failure	Nearby orbit can support shorter return if ascent and docking remain feasible.	Longer transfers increase required independent endurance in the lander.
Communications	Lunar occultation interrupts direct Earth contact from the orbiting spacecraft.	Selected NRHO offers continuous Earth line of sight; surface coverage still depends on terrain and relays.
Orbit and propulsion trade	CSM capture/departure costs support a nearby rendezvous orbit.	NRHO eases access and maintenance for the orbital asset; lander must cover the surface transfer.
Failed ascent / failed docking	Short period cannot restore a disabled ascent engine or unusable docking system.	Likewise requires an actual recovery capability; orbital infrastructure alone is not rescue.
Radiation / medical emergency	Shorter feasible evacuation helps, but shielding and emergency onset dominate.	Longer feasible evacuation increases support needs; period alone cannot determine radiation dose.

The NRHO communications and access benefits are part of NASA's orbit-selection rationale.[7] Continuous Earth line of sight from an orbital vehicle does not guarantee uninterrupted radio service from the lunar surface, or from a spacecraft with failed antennas, power, or communications equipment.

10.3 Safety Implications for Mission Design

The strongest safety distinction is the time a damaged lander must keep its crew alive before reaching a functioning return vehicle. If an emergency leaves six hours of usable life support, a nominal twelve-hour transfer is already too slow; this illustrative inequality is independent of the reliability of the docking mechanism. A recoverable engine failure with several days of cabin endurance presents a different problem from rapid depressurization or a severe fire.

NRHO operations therefore need abort trajectories sampled throughout the surface stay and orbit phase, with explicit power, cooling, oxygen, carbon-dioxide, water, navigation, and propulsion margins. Include a missed maneuver and an unsuccessful first docking attempt. Low lunar orbit requires the same analysis, including communications gaps, plane-change limits, and off-nominal ascent insertion. An Apollo-like orbit may shorten access to refuge, but using it with Orion requires a new propulsion and mass assessment; it is not a free substitution.

For the architectures studied here, Apollo has a demonstrated outbound lifeboat advantage and a flown example of relatively rapid surface-to-CSM return. Separate predeployment offers an opportunity to resolve logistics failures before crew launch, while NRHO offers communication and

orbital-access benefits. These advantages address different hazards. The evidence supports phase-specific safety comparisons, not a numerical overall crew-reliability ranking without mission-specific data.

11 Lander Dimensions, Uneven Terrain, and Tip-Over Probability

11.1 Height, Body Width, and Landing-Gear Footprint

Landing stability depends on both height and width, but the relevant ratio is center-of-mass height divided by the effective support width, not overall vehicle height divided by fuselage diameter. The support polygon is the convex hull of the load-bearing footpad contact regions. Its size and shape change when a foot has not yet touched down, a leg strokes, or a pad sinks or slides.

Lander	Overall height	Body / stowed width	Deployed landing width
Apollo LM	About 7.0 m	Ascent stage about 4.3 m across; this is not the complete lander's footprint.	About 9.4–9.5 m diagonally across landing gear.
Starship HLS	About 50 m, NASA concept reference	About 9 m parent Starship body diameter; configuration reference, not a footpad span.	Flight-configuration footpad coordinates not established by the sources used here.
Blue Moon Mk2	About 16 m (52 ft)	New Glenn's 7 m fairing is a launch-envelope reference, not a verified Mk2 body width.	Flight-configuration footpad coordinates not established by the sources used here.
Apollo-type new lander	To be sized within the spacecraft allocation	Compact two-stage packaging is proposed, not dimensionally frozen.	Size from stability and soil loads; do not inherit Apollo qualification.

Apollo dimensions are from NASA's Apollo 17 press kit; NASA supplies the modern lander height references. The 9 m Starship diameter is documented for the parent launch vehicle, and Blue Origin documents the 7 m New Glenn fairing. Neither modern width reference is a measured deployed landing-gear span. [18, 19, 20, 21, 22]

Published overall widths must therefore be accompanied by their measurement convention: body diameter, outer gear envelope, opposite footpad-center spacing, or minimum distance between opposite support edges. An outer diagonal cannot be substituted directly for the minimum support width. Nor can a fairing's outer diameter be treated as its usable internal payload clearance. No dimensions are inferred here by scaling artist's renderings.

11.2 Static Stability on a Slope

Consider a rigid lander at rest, with all intended contacts established on a planar slope, no thrust, and adequate friction. Let h be center-of-mass height normal to the support plane, and b the in-plane distance from its normal projection to the downhill tipping edge. Gravity projects the center of mass an additional $h \tan \alpha$ downhill on a slope of angle α. The static margin and limiting slope are

$$s = b - h \tan \alpha, \qquad \alpha_{\text{crit}} = \tan^{-1}(b/h).$$

Positive s is necessary for static stability about that edge; every edge must be checked. Off-center payload and asymmetric propellant inventories change b. Overall height is not h: tanks, engines,

cabin, cargo, and remaining ascent propellant determine the actual center of mass.

For ideal point contacts equally spaced on a circle of diameter D_f, a centered regular n-leg footprint has minimum edge distance

$$b_{\min} = \frac{D_f}{2}\cos(\pi/n).$$

Thus a square four-foot arrangement has $b_{\min} = D_f/(2\sqrt{2})$, rather than $D_f/2$. Actual pad areas, compliance, and missing contacts require their own geometry. Apollo's quoted outer gear span is not substituted for D_f without a dimensioned footpad drawing.

The following are *geometry-only examples*, not assignments to Apollo, Starship, or Blue Moon. They assume a centered square of point contacts.

Foot-center diagonal D_f (m)	h (m)	$b_{\min}$ (m)	Static limit (deg)
10	3	3.54	49.7
10	10	3.54	19.5
10	20	3.54	10.0
20	20	7.07	19.5

Doubling the footprint dimension offsets doubling center-of-mass height in this static ratio. It does not automatically offset the structural mass, impact energy, or soil-bearing consequences of a larger lander. Static limits are not allowable operating slopes: touchdown dynamics and uncertainty require additional margin.

11.3 Why Touchdown Can Tip a Statically Stable Lander

The first foot or pair of feet can establish a pivot while the vehicle still has sideways velocity and angular momentum. A rock can arrest a sliding pad; a depression can delay the next contact. Unequal strut compression, rebound, soil penetration, residual engine thrust, and propellant motion can then rotate the vehicle beyond its recoverable attitude. In contrast, controlled sliding or energy absorption may dissipate motion before that occurs. Sliding must be modeled, not universally credited as stabilizing.

A useful mechanics screening model considers rotation about a fixed downhill edge. On level ground the center of mass must rise by

$$\Delta z = \sqrt{b^2 + h^2} - h, \qquad E_{\text{barrier}} = m g_{\text{Moon}} \Delta z, \quad g_{\text{Moon}} \simeq 1.62 \text{ m/s}^2.$$

For the planar slope geometry above, before the static limit is reached,

$$\Delta z(\alpha) = \sqrt{b^2 + h^2} - (h\cos\alpha + b\sin\alpha).$$

Compare this barrier with post-contact rocking energy $E_{\text{rock}} = \frac{1}{2} I_e \omega_e^2$, where I_e and ω_e are inertia and angular speed about that edge. Exceeding the barrier permits overturning in the ideal fixed-pivot, lossless model; it is not a prediction for a real multi-contact lander. Contact impulses determine how translation becomes rotation, and strut work, soil work, thrust, and changing pivots alter the result.

For example, with $b = 3.5$ m and $h = 10$ m, the level-ground rise is 0.595 m. The corresponding energy per unit mass is about 0.964 J/kg, equivalent to 1.39 m/s of translation *if all of that kinetic energy were converted into overturning motion.* This is an energy-scale illustration, not a touchdown speed limit. Lower lunar gravity leaves the static slope angle unchanged but reduces the gravitational energy barrier at fixed geometry and mass.

Vertical speed also matters. With effective stopping stroke ℓ, the average kinematic deceleration scale is $v_z^2/(2\ell)$; doubling touchdown speed quadruples this contribution. Peak loads and their division among legs require force–stroke and contact models. An approximate local pressure $q_i = N_i/A_i$ must be checked against soil response for each foot, rather than assuming each receives one-quarter of the weight. Differential settlement δz across support separation L adds a tilt of approximately $\tan^{-1}(\delta z/L)$ in a two-support approximation. Plume excavation can change both the support surface and the soil conditions during descent.

11.4 Estimating the Probability of Tip-Over

A slope limit is not a probability. Define a tip-over event as unrecovered rotation causing loss of the intended upright support configuration during a specified interval from first contact through settling and engine shutdown. Leg failure, excessive settled tilt, sliding, and unsafe ascent attitude should also be reported separately, including their overlap with overturning.

Let $\mathbf{x}$ contain terrain slope and azimuth, local rock/depression geometry, soil properties, touchdown velocities, attitude and angular rates, center of mass and inertia, propellant state, leg properties, and thrust decay. For a selected vehicle V and landing-site selection policy S,

$$P_{\text{tip}}(V, S) = \int \mathbf{1}_{\text{tip}}[\mathcal{D}(V, \mathbf{x})]\, f(\mathbf{x} \mid V, S)\, d\mathbf{x}.$$

Here $\mathcal{D}$ is a validated touchdown-dynamics model, and the indicator is one for a simulated tip-over and zero otherwise. The joint distribution must preserve correlations: hazard avoidance changes the terrain actually selected, and late divert maneuvers can change touchdown velocity and attitude. Sampling independent worst-case ranges does not establish a realistic probability.

Monte Carlo sampling gives $\widehat{P}_{\text{tip}} = k/N$ for k tip-overs in N independent draws from the specified model. Report a binomial confidence interval as well as uncertainty in the terrain and vehicle models. With zero tip-overs, the one-sided 95% upper binomial bound is

$$P_{\text{upper}} = 1 - 0.05^{1/N} \simeq 3/N.$$

For illustration, zero events in 3,000 runs bounds the *modeled* probability at about 0.10%; it does not prove that the actual vehicle has that risk. Incorrect soil, contact, or guidance assumptions are not cured by more runs. No new vehicle-specific Monte Carlo campaign is claimed in this study.

Historical numerical reference. NASA's Apollo landing-gear experience report gives modeled stable-landing probabilities of 0.967 for probe-triggered shutdown, 0.998 when slopes above 12° were excluded, and 0.986 for footpad-triggered shutdown. Their complements are 3.3%, 0.2%, and 1.4%, respectively: model-specific unstable outcomes, not observed Apollo accident rates. The analysis used conservative terrain from the most severe candidate site, constrained footpads, and no crew selection of the touchdown point. These historical results demonstrate the importance of terrain selection and shutdown procedure; they are not transferable tip-over probabilities for present landers.[23]

Comparison that the evidence supports. The available dimensions do not support a defensible numerical ranking of Apollo, Starship HLS, and Blue Moon Mk2 tip-over probability. Each comparison needs deployed footpad coordinates, touchdown center of mass and inertia, validated leg/soil behavior, and the same declared terrain and operational basis. NASA's Inspector General describes an 8° settled-tilt limit for post-landing crew activities on the modern landers; that requirement is neither a tip-over angle nor a probability.[4] Hazard-detection and avoidance technology changes site selection, which must be included in the risk model. [24]

11.5 Requirements for the Proposed Apollo-Type Lander

The proposed lander carried by Apollo S-VB should be evaluated against explicit footprint and center-of-mass envelopes at touchdown, after unloading, and before ascent. Apollo S-VB itself is an Earth-departure stage and does not land on the Moon. Require stability analysis for every terrain azimuth, partial foot contact, unequal settlement, strut-stroke limits, and credible deployment failures. Validate the six-degree-of-freedom contact model against representative impact and soil tests before attaching a numerical risk claim to it.

A compact lander with a low center of mass and a wide deployed footprint is a reasonable design objective because it increases geometric margin. That benefit must be traded against leg mass, stowed packaging, ground clearance, and the area of hazard-free terrain needed. The acceptance criteria should specify both a maximum modeled tip-over probability with uncertainty bounds and separate limits for loads, settled tilt, surface operations, and safe ascent. Those risk targets remain mission-level requirements to be selected, not values inferred from Apollo's dimensions or its successful flight history.

12 Conclusion: Reconsider an Integrated Apollo-Type Lunar Stack

This study recommends a focused engineering trade of an **integrated Apollo-type upper stack on a modern launch system**: a new Apollo S-VB Earth-departure stage based on the S-IVB role, a crew command/service spacecraft, and a compact two-stage lunar lander, launched together and dispatched together toward the Moon. Here "S-IVB and upwards" includes the departure stage, its guidance and adapters, the lander, the command/service spacecraft, and the ascent protection/escape hardware. It does not mean reusing museum hardware or assuming that 1960s production and qualification can simply be restarted.

The objective is to recover three useful architectural properties: fewer independent deployment and refueling prerequisites for a landing mission; an attached, independently supplied lander available during outbound cruise; and lunar-orbit rendezvous near the surface rather than relying on a distant staging orbit for every crew return. A modern implementation would preserve independent lander batteries, life support, and pressure-fed storable propulsion, with compatible emergency interfaces and verified docked abort maneuvers. Apollo 13 demonstrates the value of that combination; it does not establish that a newly built system would inherit Apollo's demonstrated performance without testing.

The proposed new stage is named **Apollo S-VB**; this is a study designation, not an existing NASA or contractor vehicle.

The recommendation is an *architecture study*, not a finding that an Apollo upper stack is a ready-made payload for an existing rocket. The corrected Apollo 11 postflight ledger places approximately 135.1 t in its parking orbit and 134.0 t at TLI ignition. The same report records approximately 166.6 t at the first S-IVB ignition, because S-IVB also completed orbital insertion.[11] Those two launch-interface cases are very different. A carrier that completes orbital insertion can deliver a departure-only stage with less initial propellant; a booster that releases it before orbit requires an additional ascent allocation. Neither launch thrust nor nominal payload volume alone resolves this distinction.

The two candidate implementations below therefore share a common lunar mission but have different ascent systems. The Super Heavy option requires an orbital ascent stage and a newly integrated crew stack. The SLS option requires a specific ascent/upper-stage redesign or a lighter lunar stack; the published Block 1 performance is not sufficient evidence for carrying an unchanged Apollo assembly to LEO. The case for either option is strongest if it preserves a single-launch,

no-orbital-refueling mission while meeting payload, abort, landing, and return margins. If either condition fails, the claimed simplicity advantage must be recalculated rather than assumed.

13 Proposed Implementations: Apollo-Type Upper Stack on Modern Launchers

13.1 Common Lunar Stack and Mission Specification

Both concepts use the following functional stack, listed from bottom to top:

1. A new Apollo S-VB restartable LOX/LH_2 Earth-departure stage in the S-IVB class, with its own guidance, power, propellant settling, and short-duration orbital coast capability. Modern engines are permissible, but their mass and performance must be reflected in the ledger.
2. An adapter enclosing a compact two-stage lunar lander. Its descent stage carries landing propulsion and surface supplies; its ascent stage carries the crew back to low lunar orbit.
3. A command/service spacecraft with independent life support, lunar-orbit capture and Earth-return propulsion, and a command module capable of lunar-speed entry. Substituting Orion requires recomputing the lunar-orbit capture/return budget, rather than treating it as an interchangeable Apollo CSM.
4. A launch escape system and protective structure matched to the launcher, crew module, and trajectory. They may be shed before LEO, but their ascent mass and loads cannot be excluded from the launcher calculation.

The nominal sequence is one integrated launch, parking-orbit checkout, TLI, CSM transposition/docking and lander extraction, docked coast, capture into approximately 100 km low lunar orbit, lander descent, lander ascent and rendezvous, lander jettison, and CSM Earth return. An escape or protected-return trajectory must be checked at each phase. The lander lifeboat becomes available after extraction and docking, and ceases to be a complete backup after its resources are spent or it is discarded. Free-return access must be deliberately targeted; it does not follow automatically from using Apollo-shaped hardware.

For a fair first comparison, retain Apollo-like crew size, payload, and a short surface stay at an accessible site. A south-pole mission, longer stay, or a larger crew is a separate sizing case with different lighting, thermal, terrain, communications, plane-alignment, and abort requirements. This proposal does not claim that the historical equatorial Apollo trajectory meets all Artemis goals.

Illustrative departure-stage sizing. The following historical-style sizing example explains the departure-mass scale. The new Apollo S-VB requirements that follow deliberately allocate more mass to stage hardware and reserves; this example is not their final mass budget. Use 43.9 t for docked crew spacecraft plus lander, 11.2 t for stage dry hardware, 3.7 t for retained guidance/adapters, and 3.4 t for terminal propellant retained through TLI. Thus

$$m_f = 43.9 + 11.2 + 3.7 + 3.4 = 62.2 \text{ t}, \qquad m_0 = 62.2 \exp\left[\frac{3,200}{I_{sp} g_0}\right].$$

At $I_{sp} = 421$ s, $m_0 = 135.0$ t, requiring 72.8 t consumed propellant and 76.2 t initially loaded propellant including the specified remainder. At an assumed 450 s with all other masses unchanged, $m_0 = 128.4$ t. The latter is a sensitivity case, not an S-IVB engine upgrade already designed. Both neglect finite-burn corrections, coast losses, and new interface mass; they establish a first screening target, not a flight-ready mass margin.

13.2 Apollo S-VB: Preliminary Stage Requirements

Apollo S-VB is a new-build stage family. The numerical requirements below are proposed engineering targets established for this study. They are not historical S-IVB specifications, vendor commitments, or a completed design. The baseline **S-VB-D** is a departure stage delivered to orbit by an upstream ascent system. The larger **S-VB-A** combines completion of ascent with Earth departure. Stage number alone does not establish which is required: the separation state and remaining propulsion duty do.

13.2.1 S-VB-D: Baseline Earth-Departure Stage

The initial design target is 150 t for the complete assembly in a nominal 200 km circular Earth parking orbit. This includes the departure stage, retained interface hardware, and a 44 t crew-spacecraft/lander allocation. Launch escape hardware and ascent fairings jettisoned before that state belong in the ascent budget, not the 150 t orbital assembly budget.

Parameter	Proposed requirement / allocation
Role	TLI and contingency corrections after an upstream stage completes orbital insertion; no baseline orbital refueling.
Propellant / feed	LOX/LH_2, pump-fed engines; tank pressurization and restart settling provided independently.
Vacuum specific impulse	450 s design target; verify delivered performance over mixture ratio, inlet conditions, and throttle range.
Vacuum thrust	1.00 MN nominal total; one restartable, gimbaled main engine in the baseline. No main-engine-out redundancy claimed.
Stage dry allocation	15.0 t including engine, tanks, insulation, pressurization, avionics, stage power, and settling/attitude-control hardware.
Retained interface allocation	4.0 t outside stage dry mass, including load-bearing adapter, separation hardware, and retained protective structure.
Crew spacecraft + lander	44.0 t delivered through TLI; their internal propulsion and consumables are included in this payload mass.
Main propellant at TLI	87.0 t loaded; a terminal floor of 5.0 t is retained rather than credited to the required burn.
Stage wet mass at TLI	102.0 t = 15.0 t hardware + 87.0 t main propellant.
Complete TLI assembly	150.0 t = 102.0 t stage + 4.0 t interfaces + 44.0 t spacecraft.
Propulsive design duty	3.35 km/s: 3.20 km/s reference TLI plus a provisional 0.15 km/s allowance for corrections/finite-burn and operational effects.
Starts / coast	At least three qualified starts, including restart after coast; 6 h nominal parking-orbit capability and 24 h contingency design objective.
Outer envelope target	6.6 m diameter; approximately 18–20 m stage length excluding spacecraft/lander adapter. Final packaging must substantiate this target.

The 5 t terminal floor combines residuals and an unallocated contingency inventory; its usable and unusable portions must be separated by tank/feed analysis. Minor fluids and attitude-control expenditures must fit within the stated allocations or increase the departure mass. The 0.15 km/s allowance is a study choice and must be replaced by a trajectory-derived loss and reserve budget before claiming margin.

With 450 s specific impulse,

$$v_e = 4,412.99 \text{ m/s}, \quad \dot{m} = 226.6 \text{ kg/s}, \quad m_{f,\,3.35} = 150e^{-3350/4412.99} = 70.212 \text{ t}.$$

The design burn consumes 79.788 t in approximately 352 s, leaving 7.212 t of main propellant. This is 2.212 t above the specified 5 t terminal floor. Using propellant down to that floor gives

$$\Delta v_{\text{to floor}} = 4,412.99 \ln(150/68) = 3.491 \text{ km/s}.$$

The ideal margin above the 3.35 km/s design duty is therefore 0.141 km/s. At an assumed engine duty of 1.00 MN, acceleration rises from approximately 6.67 to 14.24 m/s^2 over that burn. Throttle and guidance design must address loads, pointing, and finite-burn trajectory effects at these accelerations.

Tank sizing target. Assume an oxygen-to-hydrogen mass ratio of 5.5, chosen for preliminary sizing. The 87 t load separates into 73.62 t LOX and 13.38 t LH_2. With rounded reference liquid densities of 1,141 and 70.8 kg/m^3, respectively, and a 5% tank-volume ullage allowance, minimum gross internal tank volumes are approximately 67.9 m^3 LOX and 199.0 m^3 LH_2 (266.9 m^3 combined). These are fluid-volume calculations; domes, common-bulkhead design, insulation, engine packaging, pressure limits, and fill temperature still require design. They do not by themselves validate the stage length or dry-mass allocation.

Restart, thermal, and safety requirements. Provide propellant settling, ignition sequencing, chilldown, venting, attitude control, and power through the full coast/restart envelope. Demonstrate that boiloff and conditioning losses preserve the 87 t TLI starting inventory, or increase ground loading/tank volume accordingly. The reference stage is cryogenic; the independent storable emergency propulsion resides in the CSM and LM after extraction, not in the S-VB. Provide a separation and disposal plan that avoids the crew stack and lunar assets. Qualification must cover the selected launcher's vibration, bending, staging environment, and crew-escape loads, rather than assuming inherited Saturn compatibility.

13.2.2 S-VB-A: Combined Ascent and Departure Variant

Mounting the new stage directly above Super Heavy removes the intermediate ascent stage only if S-VB itself completes the missing ascent work. A larger variant can be specified parametrically, but its final size requires the booster separation position, velocity, attitude, and ascent losses.

For a concrete preliminary case, assume a 4.50 km/s upper-stage ascent duty *including an allocated ascent-loss allowance*, followed by 3.35 km/s TLI. This is a sizing assumption, not a prediction of Super Heavy or SLS staging. Allocate 25 t stage dry mass, 4 t interfaces, 44 t spacecraft, and 8 t terminal main propellant. At an assumed constant 450 s,

$$m_{f,A} = 25 + 4 + 44 + 8 = 81 \text{ t},$$

$$m_{A,\text{LEO}} = 81e^{3350/4412.99} = 173.048 \text{ t}, \qquad m_{A,\text{start}} = 81e^{7850/4412.99} = 479.761 \text{ t}.$$

Thus the baseline D and growth A cases cannot share an unchanged tank set or dry-mass budget.

Parameter	Proposed requirement / allocation
Role	Second powered stage above Super Heavy, or a separately sized ascent/departure stage above SLS.
Assumed duty	4.50 km/s completion of ascent + 3.35 km/s TLI. Must be replaced by integrated trajectory requirements.
Stage dry / interfaces	25.0 t stage dry plus 4.0 t external retained interfaces; aggressive preliminary allocations, not validated masses.
Spacecraft allocation	44.0 t CSM + LM, as for S-VB-D.
Stage starting propellant	406.761 t, including 8.0 t retained at final cutoff; excludes any pre-ignition loss allowance.
Stage wet / total assembly	431.761 t stage wet; 479.761 t including interfaces and spacecraft at ascent-stage ignition.
At LEO insertion	173.048 t assembly; 100.048 t main propellant remains for TLI and the terminal floor.
Vacuum thrust target	4.00 MN for ascent, illustratively four 1.00 MN engines; use one engine at 1.00 MN for TLI after qualified restart.
Approximate burn durations	338 s ascent at 4.00 MN; 406 s TLI at 1.00 MN. The restarting engine sees about 744 s total nominal operation.
Engine duration / starts	Qualify at least 900 s cumulative operation for the TLI engine and at least three starts; final contingency duty may increase this.
Tank-volume scale	About 1,248 m^3 combined with the same mixture/density/ullage assumptions; about 4.68 times S-VB-D volume.
Envelope study target	9 m diameter, roughly 30–35 m stage length before spacecraft adapter; structural and packaging study required.

At the assumed ignition mass, 4.00 MN gives approximately 8.34 m/s^2 acceleration, or an Earth-weight-normalized thrust ratio of 0.85. In a moving upper-stage ascent this ratio is not a standalone pass/fail condition; the trajectory must verify that altitude, velocity, thrust direction, and burn time permit orbit insertion. The assumed 450 s vacuum performance must also be checked at the actual separation altitude. Engine-out ascent, if desired, needs a separate performance case and is not established by installing four engines.

For sensitivity, changing the assumed upper-stage ascent duty from 3.50 to 5.50 km/s changes the complete ignition mass from approximately 382.5 to 601.8 t at the same retained masses and specific impulse. This wide range is why the design must close with the actual booster before tank dimensions and engine count are frozen. The S-VB-A example ends at its specified 8 t floor; it has no additional modeled propellant margin beyond that allocation.

Build-to-spec acceptance boundary. Apollo S-VB-D is the baseline study target if the launcher can deliver 150 t to the specified parking orbit. Apollo S-VB-A is the direct-above-booster alternative, subject to a verified ascent solution. Before fabrication design is frozen, demonstrate a closed stage mass statement, achievable engine duty, thermal/pressurization performance, structural margins, trajectory and abort coverage, and manufacturable tank/engine packaging. The specifications above are sufficient to start a concept and sizing study; they are not a fabrication package or a guarantee that the assumed dry mass can be achieved.

13.3 Solution A: Super Heavy with an Orbital Ascent Stage and Apollo Top

Proposed ascent configuration: Super Heavy booster; a purpose-built orbital ascent stage; then the common Apollo S-VB-D/lander/CSM/escape assembly. Super Heavy is the first stage of

SpaceX's two-stage launch system; in the published system, Starship supplies the subsequent ascent propulsion.[16] Replacing Starship with an Apollo stack does not preserve the original system's payload-to-orbit performance.

The baseline three-stage study case uses a Starship-derived propulsion/tank stage adapted to carry the Apollo-type assembly externally above it. It is a new upper-stage configuration, not a claim that a current Starship payload bay can accommodate and safely launch a crewed Saturn upper stack. Its design must provide a structural load path, separable interface, flight control, fairing/thermal protection where required, and an unobstructed crew escape path. Booster recovery versus expendable operation is a launcher sizing variable, not a free increase in performance.

1. Super Heavy performs initial ascent and separates at the designed staging state. The added ascent stage supplies the remaining energy to reach the chosen parking orbit.
2. The ascent stage inserts and releases the 150 t Apollo S-VB-D departure assembly, then follows its own safe disposal or recovery trajectory. Retained adapter and separation equipment must fit within the validated payload budget.
3. The departure stage performs TLI after checkout, with the required propellant loaded on the ground. No tanker campaign or orbital propellant transfer is included in this reference concept.
4. The common Apollo-type sequence carries the crew and lander together to low lunar orbit and back. The Super Heavy-derived ascent hardware is not carried to the lunar surface.

Direct two-stage alternative. The S-VB-A option in Section 13.2 replaces both the intermediate ascent stage and the smaller S-VB-D. Its roughly 480 t ignition assembly is a separate booster payload/trajectory case. It preserves a single integrated launch but is substantially larger than a historical S-IVB; it cannot inherit S-VB-D's 150 t orbital mass specification.

Why not simply Super Heavy plus an unmodified S-IVB? For orientation, an idealized 421 s calculation using the historical 166.571 t first-S-IVB-ignition mass and 63.116 t final-TLI mass gives only about 4.01 km/s for both S-IVB burns combined. Allocating roughly 3.2 km/s to TLI leaves only about 0.8 km/s for completing ascent before losses. This rough allocation is not a trajectory solution, but it shows why a normally staged booster cannot be credited with delivering the Apollo top to orbit without specifying the missing ascent energy. In Saturn V, the S-II stage supplied much of that work.

Feasibility gate: demonstrate at least the selected departure-assembly mass in the specified orbit with the added ascent stage, all integration mass, and ascent/abort constraints included. Public performance for the complete Starship system is not a certification of this custom stack. The proposal preserves the intended single-launch lunar architecture, but requires major launch-system integration and a new crew-safety qualification effort.

13.4 Solution B: SLS Ascent System with an Apollo Top

Proposed ascent configuration: SLS solid boosters and core stage, followed by an Apollo S-VB carrying the common Apollo-type lander and CSM/escape assembly. Select the departure-only S-VB-D if preceding stages complete orbital insertion; otherwise resize the ascent-capable S-VB-A family for the actual SLS separation state. Study this upper stage replacing the standard upper-stage/Orion arrangement, rather than silently adding an entire fueled S-IVB on top of an already mass-limited configuration. If an extra ascent stage is necessary, include it explicitly and recompute the whole launch trajectory.

1. The SLS boosters and core accelerate the complete stack. The upper stage completes insertion if the core's separation state is suborbital.

2. Parking-orbit checkout confirms both spacecraft, docking/extraction hardware, and the departure stage. The mass at this point must close the TLI budget after all ascent propellant expenditure.

3. The departure stage restarts for TLI, after which the crew extracts and docks with the lander. The remaining lunar mission follows the common low-orbit sequence, with no baseline lunar-orbit propellant transfer or Gateway dependency.

Block 1 is not a demonstrated fit for the unchanged Apollo stack. NASA publishes about 95 t to LEO for SLS Block 1.[17] The approximately 135 t historical-style screening assembly exceeds that figure by about 40 t; the new 150 t S-VB-D design target exceeds it by 55 t. These are configuration-dependent system payload values, not a guarantee that the bare core can orbit 95 t; replacing the upper stage requires an ascent analysis. Even a hypothetical 130 t delivery capability would be slightly below the 421 s reference case before new integration margin. An assumed 450 s departure stage lowers the example to 128.4 t, but leaves little margin at 130 t and does not establish an available launcher configuration.

A complementary mass test fixes an optimistic 95 t departure assembly and solves for the supported CSM-plus-lander mass:

$$m_{\mathrm{CSM+LM,max}} = 95 \exp\left[-\frac{3,200}{421 g_0}\right] - 11.2 - 3.7 - 3.4 \approx 25.5 \text{ t}.$$

This is far below the 43.9 t Apollo reference, and below the historical fully loaded CSM-plus-LM requirement. Therefore the Block 1 path needs substantial spacecraft/stage mass reduction, a different validated launch capability, or a changed mission. Merely replacing Orion with Apollo hardware does not close this gap. Orbital assembly or refueling could alter the solution, but would change the single-launch premise and its complexity comparison.

Feasibility gate: demonstrate a particular SLS/upper-stage combination that inserts the required stack, retains the TLI propellant, meets structural and escape requirements, and supports a closed lunar landing/return ledger. Until then, Solution B is a defined design study, not a ready-to-fly SLS payload.

13.5 Selection Criteria for the Two Proposed Solutions

The first discriminator is delivered mass at the agreed parking-orbit state, not booster thrust or an advertised payload number for another configuration. Next compare the number of new stages/interfaces, the crew escape envelope, the availability of independent life support and propulsion after each failure, and the time to a reachable safe habitat. Include production restart/new-build cost, recurring launch cost, qualification flights, and operational staffing before claiming an economic advantage.

The recommended common requirement is a crew-and-lander departure in one launch without orbital refueling, with an accessible lander lifeboat during outbound cruise and a verified low-lunar-orbit abort/rendezvous plan. Super Heavy and SLS are alternative ascent foundations for testing that requirement. Neither is selected as the proven safer or cheaper implementation by the current evidence.

14 Source Treatment

Flown mission values and primary NASA/contractor specifications should be preferred. When such a value is unavailable for a developing vehicle, this working document permits Wikipedia or another secondary compilation as an explicitly identified estimate. No secondary estimate should be silently represented as a measured or finalized engineering specification. Vehicle revision and source date should accompany configuration-sensitive values. The references below support the architecture and safety discussion and the reported numerical anchors. They do not validate assumed manufacturer masses. Postflight values, published reference specifications, calculated inventories, and synthetic scenarios are distinguished explicitly. The latter close a mathematical mass budget without establishing the real vehicle configuration.

References

[1] NASA, Apollo 11 Mission Report, MSC-00171, November 1969, Table 3-I and ascent/rendezvous discussion. https://www.nasa.gov/wp-content/uploads/static/apollo50th/pdf/A11_MissionReport.pdf.

[2] NASA, Apollo 13: Mission Details, historical mission account; accessed October 11, 2026. https://www.nasa.gov/missions/apollo/apollo-13-mission-details/.

[3] NASA, Orion Reference Guide; accessed October 11, 2026. https://www.nasa.gov/reference/orion-spacecraft/.

[4] NASA Office of Inspector General, NASA's Management of the Human Landing System Contracts, IG-26-004, March 2026, pp. 5–8, 30–32 (including campaign-change addendum). https://oig.nasa.gov/wp-content/uploads/2026/03/final-report-ig-26-004-nasas-management-of-the-human-landing-system-contracts.pdf.

[5] C. Clark Esty et al., Assessment of Cislunar Staging Orbits to Support the Artemis III Lunar Surface Mission, AAS 22-762, 2022, NASA NTRS record 20220011365, Mission Risk Comparison and Table 7, pp. 16–18 of PDF. Historical design study. https://ntrs.nasa.gov/citations/20220011365.

[6] Jacob Williams et al., Targeting Cislunar Near Rectilinear Halo Orbits for Human Space Exploration, AAS 17-267, 2017, NASA NTRS record 20170001352, Return Flyby Aux Aborts. Historical design study. https://ntrs.nasa.gov/citations/20170001352.

[7] NASA, A Lunar Orbit That's Just Right for the International Gateway, July 2022; accessed October 11, 2026. https://www.nasa.gov/centers-and-facilities/johnson/lunar-near-rectilinear-halo-orbit-gateway/.

[8] Grumman, Apollo News Reference, Lunar Module Main Propulsion, pp. MP-1–MP-22, NASA-hosted manufacturer reference. https://www.nasa.gov/wp-content/uploads/static/history/alsj/LM09_Main_Propulsion_ppMP1-22.pdf.

[9] ESA, European Service Module: Propulsion; accessed October 11, 2026. https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Orion/European_Service_Module_Propulsion.

[10] A. M. Swanger et al., Large Scale Production of Densified Hydrogen to the Triple Point and Below, 2017, NASA NTRS record 20170006559. https://ntrs.nasa.gov/citations/20170006559.

[11] NASA Marshall Space Flight Center, Saturn V Launch Vehicle Flight Evaluation Report—AS-506, Apollo 11 Mission, 1969, Tables 20-5, 20-7, and 20-9 (printed pp. 20-7, 20-9, 20-12). NASA report hosted by ibiblio. https://www.ibiblio.org/apollo/Documents/lvfea-AS506-Apollo11.pdf.

[12] NASA, Artemis I Reference Guide, 2022, printed pp. 22 and 54. Reference specifications, not an as-flown event ledger. https://www.nasa.gov/wp-content/uploads/2023/03/Artemis20I20Reference20Guide_Inter.pdf.

[13] NASA, Artemis I Press Kit, November 2022, Ascent and Mission Timeline. https://www.nasa.gov/specials/artemis-i-press-kit/.

[14] NASA, Artemis I Flight Day 24: Orion Heads Home, December 9, 2022. https://www.nasa.gov/blogs/artemis/2022/12/09/artemis-i-flight-day-24-orion-heads-home/.

[15] NASA, Human Exploration and Operations Committee presentation, October 30, 2019, slide 19, Delta V Comparisons for NRHO to LLO. Historical architecture budgets. https://www.nasa.gov/wp-content/uploads/2016/05/20191030-nac-heoc-smith-v3_tagged.pdf.

[16] SpaceX, Starship launch-system description, Super Heavy first stage and Starship second stage; accessed October 11, 2026. No custom-stack performance implied. https://www.spacex.com/vehicles/starship/.

[17] NASA, SLS Reference Guide, 2022, SLS Block 1 by the Numbers; historical configuration-specific 95 t LEO reference. https://www.nasa.gov/wp-content/uploads/2022/03/sls_reference_guide_2022_print.pdf.

[18] NASA, Apollo 17 Press Kit, 1972, printed p. 99, Lunar Module dimensions. https://www.nasa.gov/wp-content/uploads/static/history/afj/ap17fj/pdf/a17_press-kit.pdf.

[19] NASA, Human Landing Systems, Starship HLS height reference; accessed October 11, 2026. https://www.nasa.gov/reference/human-landing-systems/.

[20] NASA, Industry Moon Lander Training Cabin Lands at NASA for Artemis, May 7, 2026. https://www.nasa.gov/directorates/esdmd/artemis-campaign-development-division/%20human-landing-system-program/industry-moon-lander-training-cabin-lands-at-nasa-for-artemis/.

[21] NASA, SpaceX Starship–Super Heavy Launch Vehicle at Launch Complex 39A, Final Environmental Impact Statement, Volume I, Section 2.1.2, 2026. https://netspublic.grc.nasa.gov/main/SpaceX-SSH-LC-39A-Final-EIS-Volume-I.pdf.

[22] Blue Origin, Blue Origin to Launch Telesat's Advanced Global LEO Satellite Constellation, 2019; fairing diameter reference. https://www.blueorigin.com/news/blue-origin-to-launch-telesats-advanced-global-leo-satellite-constellation.

[23] NASA, Apollo Experience Report: Lunar Module Landing Gear Subsystem, 1972, printed pp. 10–11, Statistical Landing Performance. https://ntrs.nasa.gov/api/citations/19720018253/downloads/19720018253.pdf.

[24] NASA, Safe and Precise Landing—Integrated Capabilities Evolution (SPLICE); accessed October 11, 2026. https://www.nasa.gov/safe-and-precise-landing-integrated-capabilities-evolution-splice/.